

12. Sequences and Series

12.2. Series

11.2a Learning Objectives

- Understand what a series means in relation to sequences
- Identify types of series (geometric and harmonic)
- Determine when a series converges or diverges

Suppose we have a sequence, $(a_n)_{n=1}^{\infty}$, and we want to add up all the terms. How do we sum an infinite number of values? (the short story is: very carefully, and one term at a time)

Definition 12.2.1 Series

Given a sequence of numbers (a_n) , and expression of the form

$$a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots$$

is an **infinite series**.

The sequence (s_n) is defined by

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$s_3 = a_1 + a_2 + a_3$$

$$\vdots$$

$$s_n = a_1 + a_2 + a_3 + \dots + a_n = \sum_{k=1}^n a_k$$

$$\vdots$$

is the **sequence of partial sums** of the series, the number s_n being the n^{th} **partial sum**.

If the new sequence, (s_n) , of partial sums converges to a limit L , we say that the *series* **converges** and that its sum is L . In this case, we also write,

$$a_1 + a_2 + \dots + a_n + \dots = \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \sum_{k=1}^n a_k = \sum_{k=1}^{\infty} a_k = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

Example 12.2.1

Determine the convergence of the series $\sum_{k=1}^{\infty} (-1)^k$ using partial sums.

Solution

The *sequence* this series comes from is $a_1 = -1, a_2 = 1, a_3 = -1, a_4 = 1, \dots$. This means the *sequence of partial sums* for the series is

$$\begin{aligned}
s_1 &= -1 \\
s_2 &= -1 + 1 = 0 \\
s_3 &= -1 + 1 - 1 = -1 \\
s_4 &= -1 + 1 - 1 + 1 = 0 \\
&\vdots
\end{aligned}$$

The sequence of partial sums, (s_n) , diverges, so the series, $\sum_{k=1}^{\infty} (-1)^k$, diverges.

Example 12.2.2

Determine the convergence of the **telescoping series** $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ using partial sums.

Solution

We can rewrite $\frac{1}{n(n+1)}$ with partial fractions:

$$\begin{aligned}
\frac{1}{n(n+1)} &= \frac{A}{n} + \frac{B}{n+1} \\
1 &= A(n+1) + Bn
\end{aligned}$$

So the system we get is $A + B = 0$ and $A = 1$, so $B = -1$. Then

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

This will help us write the sequence of partial sums. The n^{th} partial sum is then

$$\begin{aligned}
s_n &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\
&= \frac{1}{1} + \left(-\frac{1}{2} + \frac{1}{2}\right) + \left(-\frac{1}{3} + \frac{1}{3}\right) + \dots + \left(-\frac{1}{n} + \frac{1}{n}\right) - \frac{1}{n+1} \\
&= 1 - \frac{1}{n+1}.
\end{aligned}$$

Now we have a nice way to write the sequence of partial sums. The sequence of partial sums $(s_n)_{n=1}^{\infty} = \left(1 - \frac{1}{n+1}\right)_{n=1}^{\infty}$ converges since

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} 1 - \frac{1}{n+1} = 1$$

So the series converges to 1:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

Example 12.2.3

Determine the convergence of the **harmonic series** $\sum_{n=1}^{\infty} \frac{1}{n}$ using partial sums.

 Solution

If n is a power of 2, we get

$$s_1 = 1 = 1 + 0\left(\frac{1}{2}\right)$$

$$s_2 = 1 + \frac{1}{2} = 1 + 1\left(\frac{1}{2}\right)$$

$$s_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} > 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 + 2\left(\frac{1}{2}\right)$$

$$\begin{aligned} s_8 &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \\ &> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) \\ &= 1 + 3\left(\frac{1}{2}\right) \end{aligned}$$

This pattern continues, showing that $s_{2^n} \geq 1 + \frac{n}{2}$, so the sequence of partial sums is unbounded so it diverges. Thus, the harmonic series diverges.

11.2a Section Summary:

- We defined a series using the sequences of partial sums.
- We used the sequence of partial sums of series to discuss the convergence or divergence of some series.
- We looked at **telescoping series** and showed **the harmonic series** diverges.